

Research paper

Calculation of the energy transferred by radiation damping from nuclear spin system to receiver coil in NMR

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HIGHLIGHTS

- Find a way to calculate the energy transferred from the spin system to the detected NMR signal.
- Arrive at an analytical expression for this energy.
- Indicate the importance of radiation damping in NMR physics.

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ABSTRACT

In nuclear magnetic resonance (NMR) experiments the free induction decay (FID) current is induced by the precession of the excited magnetization, and the FID signal carries energy. This energy must come from the spin system, but has never been investigated quantitatively. This letter finds a way to calculate the energy transformed from the magnetic potential energy in the spin system to the electric kinetic energy associated with the FID current by revisiting the radiation-damping-modified Bloch equations. The result shows that under strong radiation damping conditions, almost all of the energy stored in the excited spin system is transferred to FID with negligible energy loss.

1. Introduction

Radiation damping [1–3] is an important phenomenon in nuclear magnetic resonance (NMR). It describes a pathway for the spin system to recover from the excited state to thermal equilibrium, in parallel with the relaxation pathway. When the magnetic fields applied to NMR become higher and higher [4,5], and more and more techniques are developed to enhance the sensitivity of NMR [6], radiation damping is expected to gain renewed attention in NMR [7,8]. Meanwhile, since radiation damping is related to superradiance described by Dicke [1,9], it has also caught much attention in optics and quantum mechanics [10–13].

It is well known that radiation damping causes various effects in NMR experiments, including line broadening [14], resistance to soft RF pulses [15], saw-toothed intensity profile as a function of the flip angle [16], hyperbolic dynamics of the magnetization in T_1/T_2 measurements [17,18], etc. These effects are extensively and intensively investigated both theoretically and experimentally [7,8,19,20]. However, radiation damping plays another important role in NMR, i.e., of transferring energy from the spin system to the receiver coil, which was originally pointed out by Bloembergen and Pound [1], only occasionally

mentioned in radiation damping studies [15,21,22], but has not been soundly investigated. NMR can be cataloged as “near field” optics. Energy transfer from the source (the spin system) to the destination (the FID current) is not through radio wave propagation, but through radiation damping. However, so far it has not been known how much energy is transferred through radiation damping as compared to through spin–lattice relaxation, which is known to transfer energy from the spin system to the lattice.

In this letter we perform calculations for the energy transferred through radiation damping. We arrive at an analytical expression in terms of the ratio between the radiation damping time and the transverse relaxation time. Our results show that under strong radiation damping by ignoring the contribution from relaxation to energy transfer, almost all of the energy stored in the excited spins is transferred to the receiver coil with negligible loss of energy.

2. Theory

Radiation damping has been well described by the modified Bloch equations, which was originally proposed by Bloembergen and Pound [1] and finalized by Bloom [23]:

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$$dM_{xy}/dt = -i\omega M_{xy} - M_{xy}/T_2 - (M_{xy}M_z)/(M_0 T_{RD}) \quad (1a)$$

$$dM_z/dt = -(M_z - M_0)/T_1 + M_{xy}^2/(M_0 T_{RD}) \quad (1b)$$

where T_{RD} is the radiation damping time for the tuned NMR circuit. The modification was proposed based on the law of conservation of energy [1]. Therefore, the radiation damping term in Eq. (1b) is related to energy. At any time, the energy potential of the spin system is $U = -B_0 M_z$. Before excitation, the nuclear spins are aligned with the external magnetic field B_0 , having the lowest energy potential $U = -B_0 M_0$. During excitation, the spins absorb RF energy so that their energy potential is raised up to the level of zero, and the energy stored in the spin system is $E_{spin} = B_0 M_0$. When pulse is off, relaxation and radiation damping occur simultaneously, as indicated by Eq. (1b). Since $-dU/dt = B_0 dM_z/dt$, which is the power emitted by the magnetization during Faraday induction, when multiplied by B_0 , the T_1 relaxation term is the power going to the lattice which is not interesting in this paper, and the radiation damping term in Eq. (1b) is the power going to the NMR receiver. Since M_{xy} is a function of time, the radiation damping power is a function of time as well. Then the energy transferred by radiation damping and carried by FID can be calculated by the integration

$$E_{FID} = \int_0^\infty B_0 [M_{xy}^2(t)/(M_0 T_{RD})] dt \quad (2)$$

When the expression of $M_{xy}(t)$ is known, by substituting the expression of $M_{xy}(t)$ into Eq. (2) and performing the integration, the energy transferred by radiation damping can be obtained.

In physics, the process of energy transfer is also the process of energy transform. When relaxation transforms the spin energy, which is in the form of magnetic potential, to heat, radiation damping transforms the spin energy to the electric kinetic energy of the FID current.

3. Calculations and results

In the following calculations, we assume that the spins are excited by a $\pi/2$ pulse. The simplest expression of $M_{xy}(t)$ is an exponential decay with T_2 or T_2^* as the decay constant

$$M_{xy}(t) = M_0 \exp(-t/T_2^*) \quad (3)$$

Oscillation is not considered in Eq. (3), since oscillation does not change the result of calculation. Substituting Eq. (3) into Eq. (2), we have

$$E_{FID} = (B_0 M_0 / T_{RD}) \int_0^\infty \exp(-2t/T_2^*) dt \quad (4)$$

The integration is simple and we have the result directly

$$E_{FID} = (B_0 M_0 / T_{RD}) (T_2^*/2) \quad (5)$$

or

$$E_{FID}/E_{spin} = T_2^*/2T_{RD} \quad (5')$$

However, Eq. (3) is not always valid. In the case of strong radiation damping, where relaxations can be neglected, $M_{xy}(t)$ takes a hyperbolic secant form [19]:

$$M_{xy}(t) = M_0 \operatorname{sech}(t/T_{RD}) \quad (6)$$

and Eq. (2) becomes

$$E_{FID} = (B_0 M_0 / T_{RD}) \int_0^\infty \operatorname{sech}^2(t/T_{RD}) dt \quad (7)$$

Direct integration would be difficult, however, we can find the integration indirectly. According to calculus [24] we have

$$d[\tanh(x)]/dx = \operatorname{sech}^2(x) \quad (8)$$

Integration on both sides of Eq. (8) gives

$$\int_0^\infty \operatorname{sech}^2(x) dx = [\tanh(\infty) - \tanh(0)] = 1 \quad (9)$$

Then Eq. (7) has the result

$$E_{FID} = B_0 M_0 = E_{spin} \quad (10)$$

In more general cases, $M_{xy}(t)$ is a function of time containing both parameters T_2^* and T_{RD} [20]. For simplicity, we use T_2 to replace T_2^* temporarily. When $\pi/2$ excitation is assumed, we have

$$M_{xy} = M_0 f(t) = M_0 [(T_{RD}^2 + T_2^2)^{1/2}/T_2] \operatorname{sech}[(T_{RD}^2 + T_2^2)^{1/2}(t - t_0)/(T_2 T_{RD})] \quad (11)$$

where

$$t_0 = -[T_2 T_{RD}/(T_{RD}^2 + T_2^2)^{1/2}] \tanh^{-1}[T_{RD}/(T_{RD}^2 + T_2^2)^{1/2}] \quad (12)$$

By substitution of Eq. (11) into Eq. (2) and calculating the integration, we obtain

$$E_{FID} = M_0 B_0 [1 + (T_{RD}/T_2)^2]^{1/2} - T_{RD}/T_2 \quad (13)$$

or

$$E_{FID}/E_{spin} = [1 + (T_{RD}/T_2)^2]^{1/2} - T_{RD}/T_2 \quad (13')$$

Derivation of Eq. (13) is given below. After M_{xy} being replaced by the right side of Eq. (11), Eq. (2) becomes

$$\begin{aligned} E_{FID} &= (B_0 M_0 / T_{RD}) \int_0^\infty dt [(T_{RD}^2 + T_2^2)/T_2] \operatorname{sech}^2[(T_{RD}^2 + T_2^2)^{1/2}(t - t_0)/(T_2 T_{RD})] \\ &= \frac{B_0 M_0}{T_{RD}} (T_{RD}^2 + T_2^2)^{1/2} \left(\frac{T_{RD}}{T_2} \right) \int_{-t_0}^\infty d \left[(t - t_0) \frac{(T_{RD}^2 + T_2^2)^{1/2}}{T_2 T_{RD}} \right] \operatorname{sech}^2 \left[(t - t_0) \frac{(T_{RD}^2 + T_2^2)^{1/2}}{T_2 T_{RD}} \right] \\ &= \frac{B_0 M_0}{T_2} (T_{RD}^2 + T_2^2)^{1/2} \left[\tanh(\infty) - \tanh \left(-\frac{(T_{RD}^2 + T_2^2)^{1/2}}{T_2 T_{RD}} t_0 \right) \right] \end{aligned} \quad (14)$$

Using Eq. (12) here, we have

$$\tanh \left(-\frac{(T_{RD}^2 + T_2^2)^{1/2}}{T_2 T_{RD}} t_0 \right) = [T_{RD}/(T_{RD}^2 + T_2^2)^{1/2}] \quad (15)$$

Then we arrive at Eq. (13) which is a function of the ratio between the two constants, T_{RD} and T_2 or T_2^* . The ratio of the energy transferred by radiation damping E_{FID} to the energy stored in the spin system E_{spin} can be plotted versus the ratio of these two time constants as shown in Fig. 1.

4. Discussion

When we wrote Eq. (2), we clearly meant that the energy carried by FID is the energy transferred by radiation damping. This is not an arbitrary assumption. This is a logic consequence of the law of conservation of energy, since the radiation damping theory was developed based on this law and the radiation damping feedback field is just the FID field [1]. Eq. (2) is also in agreement with Gueron [25] who concluded that the square of the FID voltage is proportional to the power of radiation damping.

Eq. (13) is the general expression of E_{FID} . In applications, T_2^* should be used to replace T_2 so that the contribution of the inhomogeneity to relaxation is included. When $T_{RD} \gg T_2^*$, Eq. (13) can be reduced to Eq. (5), and when $T_{RD} \ll T_2^*$, Eq. (13) can be reduced to Eq. (10). The approximation from Eq. (13) to Eq. (10) is quite straightforward, and the approximation from Eq. (13) to Eq. (5) is not difficult either. Eq. (13) can be rewritten as

$$E_{FID} = M_0 B_0 (T_{RD}/T_2^*) [1 + (T_2^*/T_{RD})^2]^{1/2} - T_{RD}/T_2^* \quad (16)$$

where T_2 in Eq. (13) is replaced by T_2^* . When $T_{RD} \gg T_2^*$, $(T_2^*/T_{RD})^2$ is

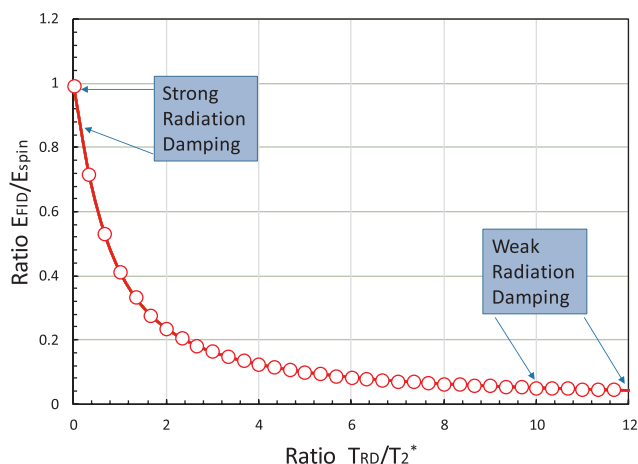


Fig. 1. Plot of the energy ratio $E_{\text{FID}}/E_{\text{spin}}$ as a function of the ratio of the two time constants T_{RD}/T_2^* . The plot shows that the energy efficiency of a NMR experiment depends on the ratio of the radiation damping time to the relaxation time. When the ratio is below 0.1, the energy efficiency can be higher than 90%, and even close to 100%, while when the ratio is above 10, the energy efficiency becomes below 5%.

a very small number and approximation after binominal expansion can be applied here: $(1+x)^{1/2} \approx 1 + (1/2)x$ when $x \ll 1$. Then Eq. (16) is ready to be reduced to Eq. (5). Since the ratio T_{RD}/T_2^* cannot be zero nor infinite, the efficiency of the transform by radiation damping cannot be 100% nor zero. The result in Eq. (10) suggests that, when radiation damping is extremely strong, i.e., when $T_{\text{RD}} \ll T_1, T_2^*$, almost all of the energy stored in the excited spins is transferred to the receiver with negligible “leakage” to the environments through relaxation. This result is in agreement with the experimental findings [17,26] that in the presence of strong radiation damping, the spin–lattice relaxation times T_1 of water can hardly be measured by inversion recovery method. Only when radiation damping is removed (by gradient or by presaturation), can T_1 be measured. Numerically, when $T_2^* = 200$ ms and $T_{\text{RD}} = 10$ ms, for example, Eq. (13') gives an efficiency $E_{\text{FID}}/E_{\text{spin}} = 95.1\%$. The leakage through relaxation is less than 5% in this case.

The result in Eq. (5), which is also an approximation of Eq. (13), suggests that, when radiation damping is weak, only a small portion of the energy is transferred to the receiver, while a big portion of the energy is wasted. The percentage of the energy transferred to the receiver is $T_2^*/2T_{\text{RD}}$, and the percentage of the energy released to the lattice is $1 - T_2^*/2T_{\text{RD}}$. In routine solution NMR, T_2^* is usually around or below 200 ms, while the radiation damping time for an interested signal (not the solvent signal) can be much longer than 10 s. Therefore, more than 99% energy is released to the lattice, and the energy efficiency is under the level of 1%. In such cases, relaxation measurement can be performed with acceptable inaccuracy, for example, by inversion recovery method, particularly for rare spins like ^{13}C , whose radiation damping time could be longer than 100 s even on a fairly high magnet and with a fairly high concentration.

Although Eq. (13) is valid for all situations with any T_{RD}/T_2^* ratio, in real applications strong radiation damping cases and weak radiation damping cases are more common than medium-level radiation damping. While strong radiation damping is encountered only when solvent suppression is a mandatory procedure for most biological samples, weak radiation damping occurs in almost all NMR experiments and also in MRI experiments. Radiation damping time for a specific rare-spin NMR signal or a tiny MRI voxel in a large coil can be as long as 100,000 s. The energy efficiency is as small as 1 ppm, but is still not zero, and FID is still detectable.

If we consider the excitation and the free precession of the magnetization as a whole, energy transform in a NMR experiment can be

simply imagined as a process of energy flow. The RF energy E_{rf} is provided by the RF pulse in the electric kinetic form. During excitation, a small portion of E_{rf} is transformed to the magnetic potential energy of the spins (E_{spin}), and most part of E_{rf} is lost in the sample, in the electronics and in the space as heat. During precession of the magnetization, E_{spin} is partly released to the lattice as heat, and partly transformed to the kinetic energy of FID (E_{FID}). Under favorite conditions, the efficiency of this E_{spin} -to- E_{FID} transform can be as high as above 95%, as seen in the above example and the curve in Fig. 1. However, in the excitation process, a NMR spectrometer performs as an extremely inefficient thermal engine. The RF pulse is usually in the range of 10 microseconds, and the RF power level is usually below 10 W. Then the RF energy supplied to the spin system is maximally at the level of 0.1 mJ. For 1 g of water, the concentration of proton is 111 M, on a 600 MHz machine, E_{spin} can be estimated to be $M_0 B_0 = 5 \times 10^{-8}$ J. Then the energy efficiency of the experiment on water is $E_{\text{FID}}/E_{\text{rf}} \approx E_{\text{spin}}/E_{\text{rf}} \approx 0.05\%$, with 99.95% of E_{rf} having been lost.

Warren et al. [15] pointed out that radiation damping is the consequence of Lenz's law, which is indeed true. According to the principles of electromagnetics [27], whenever there is Faraday induction, there is Lenz counter reaction. The law of conservation of energy guarantees that Faraday induction is always accompanied by Lenz counter reaction. Faraday's law and Lenz's law are conjugated at all times. Although Faraday induction is an active action while Lenz counter reaction is a passive action, both are essential to NMR. The resonance information is produced by Faraday induction, and the energy for the information transfer is provided by Lenz counter reaction. In this context, radiation damping is essential to all NMR experiments, which was also meant by Bloembergen and Pound when radiation damping was firstly theoretically discussed [1].

In history, Lenz's law was firstly used to determine the sign or the direction of the induced current in Faraday induction. Later it was realized that Lenz's law was consistent with the principle of conservation of energy [27,28]. However, whether in the determination of the sign of the electric current or in the interpretation of the conservation of energy, Lenz's law is always expressed in statements, but not in a mathematical equation, even when the energy involved in Lenz's law was measured [29]. Since radiation damping is so nicely described by the Bloch equations in Eqs. (1), it can be said that the Bloch equations were the first mathematical equations that quantitatively describe Lenz's law, and the description is just perfect.

When radiation damping is talked about, its bad effects [14–18], particularly the line broadening effect [14], are very much concerned. However, its bad effects can be important only when T_{RD} is comparable to T_2^* . When current technology of NMR engineering does not allow the field inhomogeneity to be less than 1 Hz (corresponding to $T_2^* = 0.318$ s), it is not necessary to worry about the bad effects caused by radiation damping when T_{RD} is longer than 10 s which is 31.4 times of the longest T_2^* . In routine NMR applications, the signals of interests have radiation damping times much longer than 100 s. There is a large space for the radiation damping times to be shortened, if NMR detection needs to be enhanced.

It should be pointed out that in this paper, no energy loss due to the resistance in the receiver circuit is considered. The resistance-caused energy loss, however, is insignificant as compared to the energy stored in the circuit. The quality factor of NMR probes, generally higher than 100, is interpreted as the ratio of the stored energy to the lost energy. Therefore, the resistance-caused energy loss is less than 1% of the total energy transferred by radiation damping.

It should also be pointed out that in this paper, it is assumed that the detection circuit is well tuned and matched. Under mistuning conditions, radiation damping involved frequency shift [7] could occur, which complicates the energy distribution and would make E_{FID} much smaller than it should be.

5. Conclusion

In summary, we have revisited the Bloch equations modified by radiation damping, and obtained an analytical expression, Eq. (13), for the energy transformed from the magnetic potential form in the spin system to the electric kinetic form in the FID current. Calculations show that the energy stored in the excited spin system can be mostly transferred to FID if T_{RD} is substantially shorter than T_2^* . The results in this letter will help to understand the physics of radiation damping better.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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